SHANMUGAMANI

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SUMMARY: Computer Vision and Machine Learning Engineer with proven experience in deploying AI-driven solutions, focusing on deep learning, computer vision, and AI applications. Demonstrated expertise in reducing process times and increasing model accuracy through innovative approaches and cross-functional collaboration.

TECHNICAL SKILLS

Programming Languages: Python, Php.
Web Technologies: HTML5, CSS, JS.
AI Topics: Data science, Image Processing, Machine Learning, Deep Learning.

Tools: Pandas, NumPy, Scikit-learn, OpenCV, Parseq, YOLO, Pyzbar

Frameworks: PyTorch, TensorFlow.

Machine Learning Topics: Generative AI, Deep Learning, Data Science, Computer Vision,
Image Processing, Multimodal Data Analysis

EXPERIENCE

Digit7 –Senior Software Engineer **12/2024- present**

- **Autonomous Retail / Computer Vision : Developed computer vision solutions for autonomous retail stores, including product identification, pose estimation, face masking, and cooler product monitoring.**
- **Improved model accuracy from 80% to 95% by optimizing vision pipelines and production workflows.**
- **Tools**: **OpenCV, SAM,Docker,Git,YOLO,**

Shrav Infotech – Computer Vision Engineer **12/2021- 11/2024**

- **Air-filter Inspection System:** Engineered a computer vision-based Air Filter Inspection System, reducing inspection time from 3 minutes to 3 seconds per unit with 95% accuracy. Integrated OpenCV with custom image processing pipelines to enhance efficiency and precision in quality control.
- **Tools**: **OpenCV, Pyzbar, Pytessract.**

- **Automatic Image Proofing System**: Developed an end-to-end Automatic Image Proofing System using OCR and QR code scanning for product ID verification. Deployed deep neural networks to cross-reference OCR, QR, and file metadata, improving verification speed from 2 minutes to 3 seconds per image with 96% OCR accuracy and 94% object detection precision.
- **Tools**: **OpenCV, Pyzbar, Parseq, YOLO, Google Collab.**

- **Engraved Text Detection System:** Designed and deployed an OCR-based system to detect and validate engraved part numbers across LH and RH components. Overcame challenges with text variability using custom pre-processing techniques. Proposed real-time deployment strategies to further optimize scalability.
- **Tools**: **OpenCV, Parseq.**

EDUCATION

- B.E Mechanical Engineering, CGPA 7.6, Dhirajlal Gandhi college of technology,**2017**, Salem.

CERTIFICATIONS

- Machine Learning Specialization -**Coursera**
- Data Science Masters Course – **PW skills**
- Web Development Course – **Accord Info Matrix**